

Arsen Khanguieldyan

arsen.khanguieldyan@gmail.com · +1 (617) 655-4650 · Boston, MA, USA

Digitalization-focused AI engineer with 7 years shipping production AI/ML and data integration systems across regulated industries (defense, automotive, enterprise). Strong on Microsoft Azure stack (AZ-400 / AZ-900 / AZ-204 certified), data architecture and ETL, low-code app delivery, and turning paper / manual workflows into structured digital systems. Recent focus: rapid AI prototypes for case intake, document understanding, and process automation.

SELECTED PROJECTS

Case Pilot — AI Surgical Case Intake & Implant Configuration

2026

scaffold-production-3492.up.railway.app · github.com/smartpneucontact-sketch/Scaffold

- Built end-to-end AI prototype: free-text surgical case description goes in, structured case record comes out with cited indication checks, recommended implant configuration, imaging-benefit rating, and MDR / FDA compliance flags.
- Agentic AI loop on Claude Sonnet 4.6 with RAG over a synthetic icotec product corpus (specs, IFU, surgical technique, prior cases, regulatory). Three custom tools: retrieve, imaging-benefit scorer, complexity classifier.
- Designed the production-target architecture explicitly mapped onto Microsoft Power Platform: Power Apps canvas UI, Power Automate flows, Azure AI Search / SharePoint as the document layer, Dataverse for implant inventory, App Insights for telemetry.
- Single-page FastAPI service with visitor-notify (Resend HTTPS API), modern dark UI, deployed to Railway.

Site Copilot — Agentic RFI & Daily Report Assistant

2026

github.com/smartpneucontact-sketch/Sufflk

- Same architecture pattern applied to a different regulated workflow (construction): two LLM agents drafting RFI responses and daily reports, with Agentic RAG and an LLMops spine (JSONL tracing, drift detector, eval gate in CI).
- Reinforces that the digitalization pattern transfers cleanly across domains — pick the workflow, pick the corpus, ship a working agent in days.

PROFESSIONAL EXPERIENCE

Hyperion

Nov 2022 – Jun 2025

Head of Engineering · Yerevan, Armenia

- Shipped a fully autonomous AI defense drone — acquired by strategic buyer; owned full engineering lifecycle across AI software, computer vision, embedded, hardware.
- Built edge-deployed computer vision for GPS-denied navigation at 96% accuracy; validated across 3,000 flight-test hours with strict regulated-industry process discipline.
- Led a multidisciplinary team (AI, CV, firmware, electronics, mechanical) and ran the full project lifecycle — requirements, design reviews, risk register, traceability matrix, verification & validation — directly applicable to medical-device process rigor.
- Designed custom PCB in Altium for flight control and ESC; ran control-loop tuning in Simulink with hardware-in-the-loop validation.

Deloitte

Apr 2021 – Jun 2022

Data Analyst · Luxembourg

- Led migration of multi-source enterprise data into a unified Azure-hosted repository with static + dynamic metadata management; exposed via REST API to a single-page web app for business-user self-service — a textbook 'eliminate manual paper / Excel workflows' project.
- Evaluated nine AutoML platforms (IaaS / PaaS / SaaS) against predefined test protocols; produced selection rationale and stakeholder presentations for client leadership.

Forschungsgesellschaft Umformtechnik mbH

May 2019 – Mar 2021

Data Engineer · Stuttgart, Germany

- Shipped a tool-wear classifier (CNN, 82% accuracy) integrated into TRUMPF Group's next-generation punching machines; co-authored a peer-reviewed publication on CNN-based tool-wear classification (Feb 2020).
- Built a sensor-fusion IoT pipeline for stamping presses (force, distance, sound, lubrication) and an ML failure-prediction algorithm on Audi AG data — connecting operational sensor data into actionable signals, the data-integration pattern icotec's JD asks for.

Audi AG

Apr 2018 – Dec 2018

Software Developer · Neckarsulm, Germany

- Built and deployed an Oracle APEX low-code web application that digitalized die-cast tooling improvements — exactly the kind of paper-to-digital workflow this role targets. Rolled out internationally across Audi sites.

TECHNICAL SKILLS

Digitalization & Low-Code

Oracle APEX (production), Microsoft Power Platform-shaped architecture (Power Apps, Power Automate, Dataverse, AI Builder — applied via prototype), rapid digital-form delivery, process automation

Microsoft / Azure

Azure (AZ-400, AZ-900, AZ-204 certified), Azure Data Factory, App Insights-shaped telemetry, Active Directory, REST API design, custom connectors

AI & ML

LLMs (Claude, GPT, Azure OpenAI), Agentic RAG, tool-use, prompt engineering, PyTorch, HuggingFace, YOLO, computer vision, CUDA, LLMOps

Data Architecture & Integration

ETL design, sensor-fusion pipelines, REST APIs, vector stores, hybrid search (BM25 + dense), Azure AI Search-shaped retrieval, AWS, Docker, Kubernetes, Terraform, GitHub Actions, SQL

Electronics & Hardware

Altium / PCB design, Arduino, sensor integration, SolidWorks, Simulink — regulated industry V&V and documentation discipline

Soft skills

Stakeholder workshops, requirements engineering, training & enablement, cross-functional team leadership, multilingual (EN/FR/DE/HY)

CERTIFICATIONS

Microsoft Azure DevOps Engineer Expert (AZ-400) · Microsoft Azure Administrator (AZ-104 / AZ-204) · Microsoft Azure Fundamentals (AZ-900) · AWS Cloud Practitioner · TensorFlow Developer · Deep Learning in Computer Vision

EDUCATION

Ecole Centrale d'Electronique — Paris, France

M.S. Computer Science & Engineering	Jun 2019
B.S. Mathematics & Electronics	Jun 2017

MIT Professional Education

Digital Transformation in the AI Age (in progress)	Dec 2025 – Sep 2026
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LANGUAGES

English (proficient) · French (fluent) · German (fluent) · Armenian (native)